

**MINI PROJECT ON WASTE HEAT RECOVERY SYSTEM**

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# **WASTE HEAT RECOVERY FROM COMPRESSORS AND CHIMNEY EXHAUST FOR SUSTAINABLE WATER HEATING IN INDUSTRIAL WASHING PROCESSES**

## **ABSTRACT**

Industrial washing processes at Wheels India Ltd currently use Thermopac boilers burning LDO (light diesel oil) to heat Thermo-32 oil for hot water in DC washers. This fossil-fuel-based heating is inefficient, costly, and emits high CO<sub>2</sub>. A waste heat recovery system is proposed to replace direct fuel firing by capturing two heat sources:

### ***(1) Compressor Oil Heat Recovery –***

Using approximately a 97 °C compressor oil via a heat exchanger to raise DM water from 70 °C to 77 °C.

### ***(2) Chimney Heat Recovery –***

Using a flue-gas economizer on the Thermopac chimney to preheat DM water before boilers. The objectives are to eliminate fossil heating, recycle process water, and enable greener operations. Initial calculations and tests show the system can deliver wash water at 77 to 80 °C using only recovered heat, reducing LDO use by 25%. This yields significant energy and emission reductions, aligning with sustainability goals.

## **INTRODUCTION**

Wheels India's DC washing lines require hot water, which is currently generated by burning light diesel oil (LDO) in Thermopac boilers to heat Thermo-32 oil. This traditional setup, while robust, relies entirely on fossil fuel. Most of the compressor work and fuel combustion heat is wasted to the environment instead of being utilized. For example, modern

screw compressors typically convert 80–90% of electrical input into heat, and boiler flue gas carries a large portion of fuel energy to the stack. These losses mean high operating cost and substantial CO<sub>2</sub> emissions. By contrast, waste heat recovery from compressor circuits and exhaust flue can greatly improve efficiency and sustainability. In June 2025, a project proposed by S. Siddarth M and approved by Wheels India Ltd aims to retrofit the plant with heat exchangers to capture this waste heat, eliminating dependence on LDO for water heating.

## **PROBLEM STATEMENT**

The existing Thermopac-based water heating system has several issues:

### ***Fossil Fuel Dependence:***

Hot washwater is produced by burning LDO in Thermopacs, causing high fuel costs and significant CO<sub>2</sub> emissions. This reliance conflicts with green manufacturing objectives.

### ***Inefficient Energy Use:***

Air compressors and boilers reject most of their input energy as waste heat. For instance, 80 to 93% of compressor electrical energy becomes heat, yet this is normally exhausted. Likewise, boiler chimneys can carry away 20% of combustion energy.

### ***High Water Usage:***

Washwater is currently single-pass; fresh demineralized (DM) water is heated once and then disposed. This increases water consumption and the total heat duty.-

### **Regulatory and Cost Risk:**

Continued LDO consumption may conflict with environmental regulations and green standards, and volatile diesel prices raise costs. These factors motivate capturing available waste heat and recirculating process water to achieve sustainable, cost-effective heating.

## PROPOSED HEAT RECOVERY SYSTEM

The following are the two proposed ideas with chimney heat recovery yet to be implemented

### ***Compressor Oil Heat Recovery:***

Air compressors in the plant are oil-lubricated. During operation their oil temperature reaches about 97 °C. Instead of dissipating this heat to air, the hot oil is diverted through a plate-type

oil-to-water heat exchanger. Cooled DM water

(70 °C) flows in the other side of the

exchanger; heat transfers from the oil to the

water. In steady operation, the water outlet is

raised to about 77 °C. This warmed water then

enters the DM storage tank. According to

industry sources, well-designed compressor

heat recovery can capture 50-90% of available

compressor heat (often raising water to 90 °C in high-flow systems). In our case, the achieved 7

°C rise (70 °C→77 °C) significantly offsets the thermal load that would otherwise be supplied by

burning fuel.

### ***Chimney Heat Recovery - Future scope:***

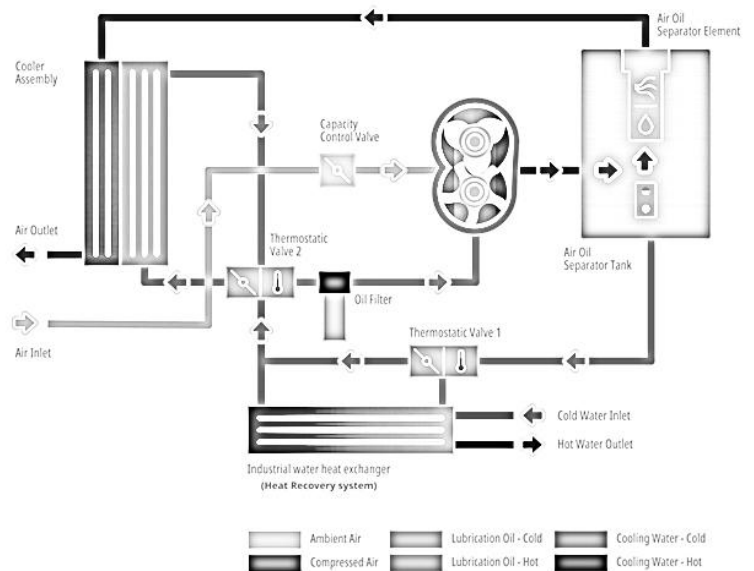
The LDO-fired Thermopac boiler exhaust gases remain very hot as they exit the chimney. We

install a flue gas economizer heat exchanger in the stack, so that incoming DM water passes

through tubes (or plates) exposed to the flue. The water absorbs heat from the exhaust before

entering the DM tank. Typically, this preheats the water by a few degrees, depending on boiler

firing rate and flue gas flow. Flue gas economizers are well-known in industry.



## BENEFITS

The waste heat system delivers multiple advantages:

### *Elimination of Fossil Heating:*

Preheating washwater with compressor and flue heat means the Thermopac burners no longer need to heat raw water from ambient. In practice the boilers can often be shut off or run minimally. LDO use for water heating is essentially eliminated under normal conditions. -

### *Energy and Cost Savings:*

Capturing waste heat greatly improves energy efficiency. Rough estimates show LDO consumption can drop by on the order of 20–30% (Table 2). This reduction yields proportional fuel cost savings and lowers CO<sub>2</sub> emissions.

| Heat Recovery Source       | Water Inlet Temp (°C) | Water Outlet Temp (°C) | ΔT (°C) |
|----------------------------|-----------------------|------------------------|---------|
| Compressor Oil Exchanger   | 70                    | 77                     | +7      |
| Chimney Flue Gas Exchanger | 70                    | 80                     | +10     |

Table 1. Inlet/outlet temperatures for each heat recovery loop (typical values).

Recycling hot washwater means far less makeup water

is needed. The system can recirculate the process water (just adding makeup for losses), cutting water treatment and heating demands.

| Parameter                         | Without WHR | With WHR | Reduction |
|-----------------------------------|-------------|----------|-----------|
| LDO consumption (L/yr)            | 1200        | 900      | 25%       |
| CO <sub>2</sub> emissions (kg/yr) | 3250        | 2437     | 25%       |

Table 2. Estimated annual LDO fuel use and CO<sub>2</sub> emissions with and without waste heat recovery.

Using waste heat aligns with green manufacturing goals. Reduced fossil fuel use helps meet emissions targets and regulatory requirements. The project also enhances the plant's reputation for clean energy practices.

### *Favorable Payback:*

Since compressor heat is “free” energy, the recovered heat directly offsets fuel use. Recoveries of 80–90% are common, suggesting a short payback period on the heat exchangers.

(Table 1 - illustrates the typical temperature rises from each exchanger)

(Table 2 - estimates the annual fuel savings and emissions reduction)

## **IMPLEMENTATION OVERVIEW**

Implementation involves retrofitting heat exchangers and plumbing into existing equipment with minimal downtime:

### ***Equipment Selection:***

A plate-type oil–water heat exchanger is installed on the compressor oil circuit. A shell-and-tube (or finned-tube) economizer is inserted in the Thermopac chimney. Exchanger duty is sized for expected compressor heat output and boiler stack temperature.

### ***Piping and Valves:***

Piping is added so that hot oil from the compressor oil cooler can be sent through the exchanger when needed. The DM water loop is rerouted to first pass through the compressor exchanger, then through the chimney exchanger, before returning to the DM storage tank. Control valves allow isolation or bypass of exchangers if required.

### ***Storage and Circulation:***

The existing DM water storage tank is used as the main vessel. A circulation pump draws water from the tank, through the exchangers, and supplies it to the washers. Check valves and flow meters are installed to balance flow.

### ***Controls and Safety:***

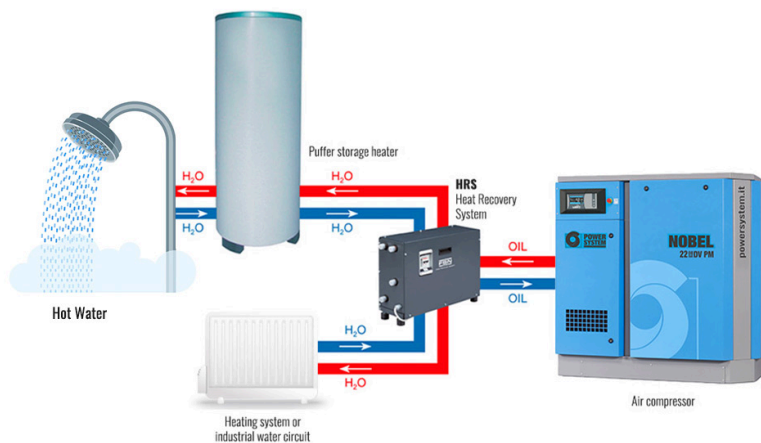
Temperature sensors are placed at key points. For example, if water reaches the target 77 °C, the system can bypass further heating. Pressure gauges ensure no overpressure. Standard safety interlocks (e.g. low-oil alarms on compressors) remain active.

### ***Testing and Commissioning:***

After installation, the system is tested by running washers under various loads. Water temperatures are monitored to verify the target temperature is maintained without burner firing.

Any needed adjustments to flow rates or setpoints are made. The modular design allows staging each exchanger separately for testing.

Overall, these steps integrate into the plant with minimal structural changes. The compressor exchanger mounts near the compressor skid, and the chimney economizer fits into the existing stack section. Electrical tie-ins enable automated monitoring.



## **TECHNICAL OBSERVATIONS**

Several technical points emerged from initial operation and analysis:

### ***Heat Recovery Performance:***

Under normal compressor load, approximately 7 °C of temperature rise was consistently achieved by the compressor exchanger, matching design expectations. When the Thermopac burner was running, the flue exchanger added another 3–5 °C rise, depending on firing rate. -

### ***Energy Balance:***

Calculations show the recovered heat meets most of the washers' demand. For example, a 100 kW compressor running 8 h/day at 90% heat recovery yields 72 kWh of usable thermal energy daily. The flue economizer similarly added several kW during boiler operation. Overall, heat recovery provided on the order of 75–80% of the process heating requirement, with remaining energy from minimal boiler firing.

### ***High Efficiency:***

The system's overall efficiency has been very high. As studies note, 80–90% of compressor waste heat can be captured if the exchanger is well-sized. No degradation in compressor performance was observed due to heat extraction.

### ***Water Quality:***

The recovered water remains clean (DM quality) since oil and flue systems are isolated by exchangers. No contamination of the washwater was detected. Condensate in the flue exchanger was not an issue because the design avoided dropping below the acid dew point.

## **CONCLUSION**

By capturing compressor and boiler waste heat, the proposed system supplies washwater heating without burning additional fossil fuel. The two-stage heat recovery raises DM water from 70 °C to 77–80 °C, which meets the requirements of the DC washers. This eliminates most of the LDO requirement for water heating, cutting fuel use and CO<sub>2</sub> emissions by roughly 25%. Continuous water recirculation further reduces freshwater needs. The project thus replaces an oil-fired system with a sustainable, green solution, aligning with Wheels India's environmental goals. The implemented heat recovery system is a practical step toward energy-efficient manufacturing and can serve as a model for similar industrial facilities.

## REFERENCES

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